C4 Ingestion and orchestration

Data Factory pipelines, Dataflows Gen2 and copy jobs.

C4-L1 Choosing how data gets into Fabric

Mirroring for curated databases, Copy job for bronze, Eventstream for events, Spark or Dataflow Gen2 for transformation, and pipelines to orchestrate.

Microsoft's starting rule by layer	Gold from existing systems: Mirroring. Bronze ingestion: Copy job. Low-latency events: Eventstream.
What a pipeline is for	Orchestration: ordering, branching, loops, retries and alerts across activities.
Spark or Dataflow Gen2?	Spark for code-first, large-volume engineering. Dataflow Gen2 for low-code Power Query authors.
Copy activity vs Copy job for incremental loads	Copy activity needs hand-built state. Copy job tracks the last successful run itself.

C4-L2 Incremental copy with a watermark, or with CDC

Watermarks copy rows above the last value seen: no deletes, no late rows. Use CDC or a ROWVERSION column, and Merge for one row per key.

Copy job incremental copy	First run copies everything. Later runs copy new or changed data. State is tracked by the job.
Watermark blind spots	Cannot see deletes. Misses late rows at or below the stored watermark.
When to use CDC	When deletes must replicate, or changes arrive out of order.
Copy job update methods	Append (default), Merge, Overwrite, SCD Type 2. Merge for one row per key.

C4-L3 Pipelines that fail fast, retry sensibly and alert

Set timeouts near the real run time, retry only transient failures, chain on Upon Success and alert on Upon Failure.

Activity timeout default	12 hours, maximum seven days, format D.HH:MM:SS. Set it near the real run time.
Activity retries	Only when enabled. Count defaults to 1, interval 30 seconds. Retry transient failures only.
Dependency branches	Upon Success, Upon Failure, Upon Completion (either way), Upon Skip.
@{...} vs @...	@{...} inside text always yields a string. A bare @ expression keeps the value's type.